Khaled Waleed Ba Tais

AI & Computer Vision Engineer | Edge AI

+966 55 655 7159 | khaled0wleed@gmail.com | Kuala Lumpur | **LinkedIn** | **GitHub** | **ORCID** | **khh4lid.github.io/portfolio**

SUMMARY

Third-year Computer Science (AI) student who ships computer vision on real hardware, not just notebooks: field data collection and annotation, training with PyTorch and YOLO, and real-time on-device inference on Raspberry Pi 5, NVIDIA Jetson and Hailo-8. First author of the DragonFruit-MY dataset preprint, Darb 2026 winner as team leader, and Top 25 of 225+ teams at GSR Hackathon 2026 (KFUPM). Seeking an AI, Computer Vision or Edge AI internship, **19 Apr to 6 Aug 2027**.

EDUCATION

Taylor's University, Malaysia, dual award with the University of the West of England, Bristol — Sep 2024 to 2027
BSc (Hons) Computer Science (Artificial Intelligence) | **CGPA 3.73 / 4.00** | Focus: deep learning, computer vision, embedded AI
Dean's List, 5 consecutive semesters (Feb 2025 to Apr 2026)

TECHNICAL SKILLS

AI / ML: Python, PyTorch, TensorFlow/Keras, scikit-learn, Random Forest, XGBoost, NumPy, Pandas
Computer Vision: YOLOv5 / YOLOv8, CNNs, OpenCV, MediaPipe, object detection, segmentation, tracking
Edge AI: Raspberry Pi 5, NVIDIA Jetson, Hailo-8 (26 TOPS), ONNX, INT8 quantization, on-device / offline inference
Embedded & IoT: ESP32, Arduino, sensors, MQTT pipelines, 3D printing / CAD enclosures
Tools: Git & GitHub, Linux, Docker, REST APIs, Hugging Face Spaces, Vercel, Google Colab

PROJECTS

AquaGuard | Edge AI Drowning Early Warning — Live | Code

- Built a fully on-device drowning detection system on YOLO, Raspberry Pi 5 and Hailo-8, with no cloud in the alert path.
- Trained on **8,689 images**: **90.5% mAP50** for person detection and **94.2% mAP50** for pool-zone segmentation.
- SAFE / DANGER / EMERGENCY decision logic with Telegram alerts in **under 2 seconds**; validated in a real pool test.

Tech: Python, PyTorch, YOLO, OpenCV, Raspberry Pi 5, Hailo-8, edge deployment

ClimaShield | Edge AI Smart Agriculture — Video | Live | Code

- Autonomous rooftop and greenhouse guardian: camera and sensor data feed on-device models (Random Forest, XGBoost) that decide when to shade and irrigate, fully offline.
- YOLOv8n leaf screening (9 classes) at **86.4% mAP50**, about **175 ms** on Raspberry Pi 5; ESP32/MQTT sensors, ESP32-S3 smartwatch, web dashboard, grow-rig prototype.
- **Top 25 of 225+ teams** at GSR Hackathon 2026 (KFUPM) and **Top 33 of 439 teams** at the Asian Hackathon for Green Future 2026.

Tech: Python, PyTorch, OpenCV, scikit-learn, Raspberry Pi, ESP32, MQTT

SignBridge | Two-Way Arabic Sign Language Translator — Demo | Live | Code

- Built a real-time, two-way translator: Arabic sign language to speech, and speech back to text, for Deaf and hearing users.
- Shipped a public interactive demo on Hugging Face Spaces and a web app on Vercel; **winner of the Darb 2026 hackathon**.

Tech: Python, PyTorch, OpenCV, MediaPipe, Hugging Face

RESEARCH

- **DragonFruit-MY** (first author, with Dr Husna Sarirah Husin; SSRN preprint, 2026): open dataset of 727 dragon fruits photographed in the field, each labelled for disease and ripeness; YOLOv8n baseline mAP@0.5 0.966. **Preprint** | **Dataset**
- **Paper presenter, 9th Taylor's International SEARCH Conference 2026:** "Algorithmic Translation at the Edge: SignBridge as a Generative Media Conduit for the Arabic Deaf and Hard-of-Hearing Community," Oct 2026.

AWARDS & RECOGNITION

- **Darb 2026: 1st place, Digital Education track (MYR 2,000)** with SignBridge; honoured by the Ambassador of Yemen.
- **GSR Hackathon 2026, KFUPM: Top 25 of 225+ teams**, Sustainable Agriculture track, with ClimaShield (Oct 2026).
- **Asian Hackathon for Green Future 2026: Top 33 of 439 teams** from 22 countries (For Green Future Foundation, VinUniversity).
- **MEWA University Innovation Challenge** (2nd edition, Saudi Arabia): qualified for the next stage (Oct 2026).
- **MyIPO copyright notification** CRLY2026W05575 for the ClimaShield system architecture and operating flow (July 2026).

CERTIFICATIONS

Claude Academy (Anthropic): Building with the Claude API (Oct 2026) | **Google:** Gemini Certified Student (Feb 2026) | **Cisco Networking Academy:** IoT, Networking and Packet Tracer courses (Nov 2025) | **Salesforce:** Reports & Dashboards Superbadge (Nov 2024)

LEADERSHIP & ACTIVITIES

- **Team leader:** led teams that shipped under hackathon deadlines, including **MedSeek**, an AI medicine safety checker (SAFE / CAUTION / NOT SAFE) built in one day at Lovable Vibeathon KL 2026 (AWS office), and **KopiKita**, built in two days at RMIT Hackathon 2026.
- **Industry:** Student Research Assistant for Taylor's at **AI Expo Korea 2026** (Seoul); interviewed by Estidamah at GSR 2026; **TV interview** at ENERtec Asia 2026; presented ClimaShield to TNEX; ITEX 2026 and LEAP 2026.